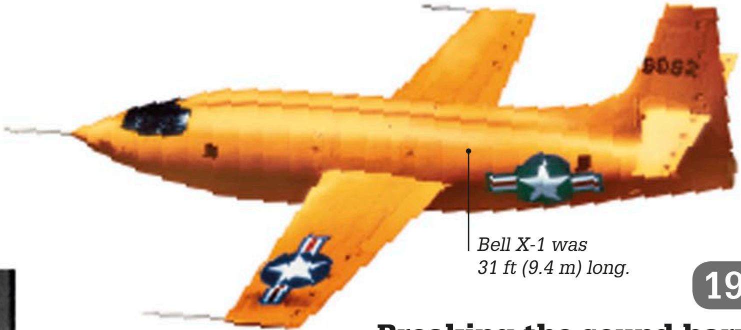
1945 ▸ 1950

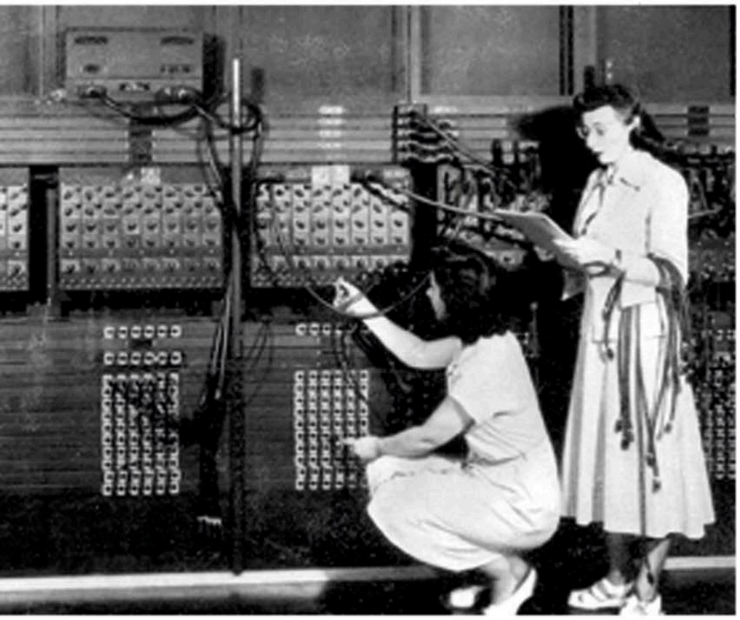


Bell X-1 was 31 ft (9.4 m) long.

1947

Breaking the sound barrier

US Air Force pilot Charles "Chuck" Yeager flew the rocket-powered Bell X-1 aircraft at an altitude of about 43,000 ft (13,000 m) to become the first human to fly faster than the speed of sound, which is 660 mph (1,062 km/h) at this height. The X-1 has a top speed of 700 mph (1,130 km/h).



Programmers reprogram ENIAC using cables plugged into boards

1946

ENIAC operational

ENIAC (Electronic Numerical Integrator And Computer), the first electronic, general-purpose computer, started operating at the University of Pennsylvania. This 98-ft- (30-m-) long machine weighed more than 60,000 lb (27,000 kg) and ran different applications, from weather forecasts to calculating the impact of nuclear bomb simulations, until 1955.

1947

Discovery of promethium

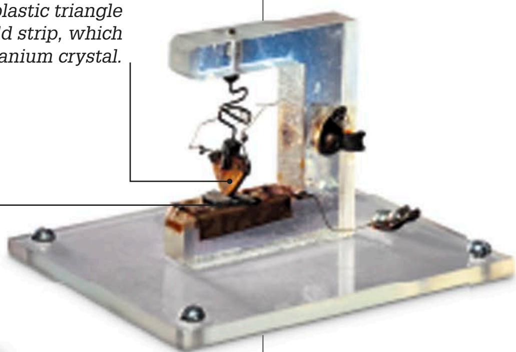
A gap in the periodic table (see pp.188–189) was filled with the discovery of the missing element (with atomic number 61) by chemists at Oak Ridge National Laboratory in the US. It was named promethium (Pm).

1945

1946

Bouncing off the Moon

The United States Army Signal Corps (USASC) was the first to bounce radio waves off the Moon. Named Diana after the Roman goddess of the Moon, the USASC's project used World War II radar antennae to send the signals and receive them back from the Moon 2.5 seconds later.



Weak electric signal enters via one side of the plastic triangle covered in gold strip, which touches a germanium crystal.

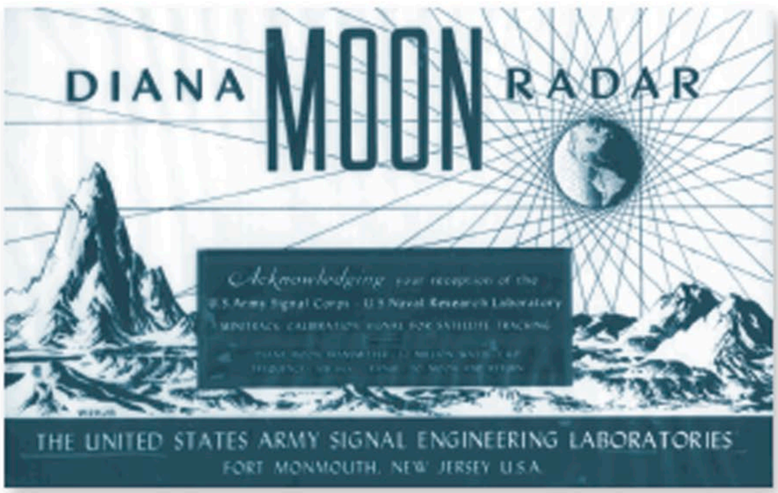
Germanium crystal, on a metal base, amplifies the weak electric signal and makes it stronger.

Replica of Bell Labs' original transistor

1947

First transistor

The transistor was invented by American physicists William Bradford Shockley, Walter Houser Brattain, and John Bardeen at Bell Labs in New Jersey. This electronic component could act as a switch in electric circuits, or as an amplifier, increasing the strength of an electric signal. Transistors replaced bulky vacuum tubes and allowed smaller, faster, cheaper, and more reliable electronic goods to be made.



Each person who heard the returned Diana signal received this keepsake.

In 1947, British engineer Dennis Gabor invented holography—a system of displaying holograms (three-dimensional images on a two-dimensional surface).